SARTHAK SHRIVASTAVA

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EDUCATION

Indian Institute of Technology Delhi — India
B.Tech, Civil Engineering, GPA 8.283/10.0 — *2019 - 2023*

SCHOLASTIC ACHIEVEMENTS

- Awarded the **Summer Undergraduate Research Award (SURA), 2021** from Industrial R&D Unit, IIT Delhi — *2022*
- **Online Judges:** Scored **5-star rating** on ***HackerRank*** and **3-star (max. 1783 rating)** on ***CodeChef*** — *2021*
- **JEE (Advanced):** Secured **All India Rank 1841 (top 1 percentile)** among 0.17 million candidates — *2019*
- **JEE (Main):** Achieved **All India Rank 1403 (top 0.1 percentile)** among 1.2 million candidates — *2019*

PROJECTS

Peridynamics-Based Modeling of Nanoindentation Behavior of Silicate Glass (*Sep, 2022 - Present*)
Advisor: Dr. N. M. Anoop Krishnan, IIT Delhi

- Developing a **peridynamics**-based numerical model to investigate the nanoindentation behavior of silicate glass under **varying tip angles of the indenter**, providing insights into the **microscale** mechanical behavior of the material
- Validating the numerical model using **finite element analysis** (FEA) software **ABAQUS**, demonstrating the accuracy and reliability of the proposed model in predicting the **nanoindentation response** of silicate glass
- Conducting **upscaling analysis** to investigate the **macroscopic** behavior of silicate glass under different nanoindentation conditions, providing a better understanding of the mechanical properties of the material at the macroscale level

Stability of Steel Columns at Elevated Temperatures | *Finite Element Analysis* (*Nov, 2022 - Dec, 2022*)
Advisor: Dr. Vasant Matsagar, IIT Delhi

- Eigen value problem to determine the **buckling load** through FE formulation showed convergence with Euler's formula
- Stresses were evaluated under **thermo-mechanical loading** for various boundary conditions of the column
- Performed **non-linear analysis** of columns using **Modified Riks approach** in **ABAQUS**; determined **critical thermo-mechanical stresses** and facilitated the performance of **post-buckling analysis** until collapse

The Application of Machine Learning to Structural Health Monitoring | *SURA* (*May, 2021 - Oct, 2021*)
Advisor: Dr. Sahil Bansal, IIT Delhi

- Devised a **vibration-based damage-detection** approach using **AI methods** on time response simulated data
- Extracted different **damage-sensitive features** based on **signal statistics** and **modal properties**; analyzed **signal response time-series** using **AR and ARX models**; proposed **statistical model for feature discrimination**
- Ranked damage-sensitive features and employed **support vector machines** to classify **structural condition states**

Transverse Vibration Analysis of Euler-Bernoulli Beam | *Computational Mechanics, Structural Dynamics* (*Mar, 2022*)
Advisor: Dr. Prapanch Nair, IIT Delhi

- Derived an **explicit finite difference scheme** with **second order accuracy** by discretizing the governing equations
- Implemented Python script for computation of **natural frequencies** of vibration, **mode shapes**, and **vibration response** of the elastic beam for **different boundary conditions**, viz. fixed-fixed, fixed-pinned, pinned-pinned, and fixed-free
- **von Neumann analysis** was done to establish **stability requirements** of the solution, explicit FD scheme was found to be **conditionally stable** and **convergent**; compared the results with analytical solution to validate the approach

Structural Design using STAAD.Pro | *Structural Design & Detailing* (*Sep, 2022 - Nov, 2022*)

- Designed and analyzed the structures of **hostel campus** comprising five blocks of **G+10 storey buildings**, **reinforced concrete overhead water tank**, **industrial shed truss**, and **transmission tower** using AutoCAD and STAAD.Pro
- Resistance to **earthquake** and **wind loads** were incorporated into the structural design process as per Indian Standards

Food Outlet Management System | *Data Structures & Algorithms* (*Apr, 2021*)

- Developed a simulation environment for a food outlet having facility of **multiple ordering/billing counters** and food delivery on **first-order-first-serve basis**; implemented **Priority Queue**, **AVL Tree**, **Array Deque** based data structures
- Statistics like **average waiting time**, **average queue length** were computed in order to **improve customer service**

RELEVANT COURSEWORK

Calculus, Linear Algebra & Differential Equations, Engineering Mechanics, Mechanics of Solids, Mechanics of Fluids, Building Materials, Structural Analysis – I, Structural Analysis – II, Geotechnical Engineering, Reinforced Concrete Design, Steel Design, Structural Dynamics, Structural Design & Detailing, Data Structures & Algorithms, Computational Mechanics, Advanced Mechanics of Solids, Finite Element Methods in Structural Engineering, B.Tech Project - I

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Java, Julia
- **Software & Tools:** Microsoft Office, Visual Studio Code, STAAD.Pro, Abaqus FEA, MATLAB, LaTeX, GitHub
- **Libraries:** C++ STL, Pandas, NumPy, SciPy, Matplotlib, Scikit-learn, Selenium, Tesseract-OCR, OpenCV

TEACHING & MENTORING EXPERIENCE

Course Hero, Inc. | *Tutor* (*Dec, 2021 - Present*)

- Offered online tutoring to students seeking academic assistance in Physics, Mathematics, and Civil Engineering
- Received **consistently positive ratings** and feedback from students for effectiveness and clarity of instruction

Academic Mentor | *Board of Student Welfare (BSW), IIT Delhi* (*Nov, 2020 - Feb, 2021*)

- Provided **peer mentorship** and tutoring to undergraduate freshers in **Engineering Mechanics** course
- Assisted the students to acclimate to a new academic setting by **facilitating weekly study sessions**

Unacademy | *Educator (IIT-JEE Physics)* (*Aug, 2020 - Oct, 2020*)

- Taught Advanced Physics to **500+ IIT-JEE aspirants**; held interactive sessions and emphasized **analytical thinking**, **problem-solving** and **exam-taking** skills through the use of **visual learning** approaches; offered learning resources
- Ranked among **top 100 educators** in IIT-JEE category with over **15K minutes** of watch time on online platforms

Project Aarohan | *National Service Scheme (NSS), IIT Delhi* (*Sep, 2019 - Apr, 2020*)

- Mentored underprivileged students of class XI and XII for competitive exams
- Taught Physics; provided them with comprehensive study material and conducted doubt clearing sessions

WORK EXPERIENCE

Axis Bank | *Summer Analyst (Business Intelligence Unit)* (*Jun, 2022 - Jul, 2022*)

- Built a **web-scraping tool** that extracts information related to employees' salaries, revenues, capitals of an organization
- Employed **Selenium** to **automate web browser** and interact with **dynamic web pages**; handled **captcha recognition** through **OpenCV** based **character segmentation** technique; analyzed the scraped data to gain useful **business insights**
- Developed tool is capable of scraping data of **600K+ organizations**; proposed ideas to **amplify sales** of salary accounts